GEOGRAPHY 02 — THE EARTH'S CRUST, ROCKS / INDIA GEOLOGICAL STRUCTURE

CRUST, LITHOSPHERE AND THE → IGNEOUS, SEDIMENTARY AND METAMORPHIC → FROM CONTINENTAL DRIFT TO → INDIAN PLATE JOURNEY AND → INDIAN ROCK SYSTEMS, BASINS → HIMALAYA, FORELAND, COASTS, ISLANDS

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g11 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 CRUST, LITHOSPHERE AND THE ROCK-CYCLE NETWORK

CRUST, LITHOSPHERE AND THE ROCK-CYCLE NETWORK | EARTH MATERIAL SYSTEM MINERAL | COMPOSITIONAL SHELL | CRUST + RIGID UPPERMOST MANTLE | WEAKER MECHANICAL LAYER BELOW | ROCK-CYCLE NETWORK MAGMA --CRYSTALLISATION-→ IGNEOUS WEATHERING + TRANSPORT | <---- MELTING -- METAMORPHIC <-- SEDIMENTARY HEAT + PRESSURE | THE CYCLE IS A NETWORK

EARTH MATERIAL SYSTEM mineral → rock → lithology → petrology → geology → Crust = compositional shell → Lithosphere = crust + rigid uppermost mantle → Asthenosphere = weaker mechanical layer below → ROCK-CYCLE NETWORK magma --crystallisation-→ IGNEOUS weathering + transport

DEFINITION / COMPONENTS

- EARTH MATERIAL SYSTEM mineral → rock → lithology → petrology → geology
- Crust = compositional shell
- Lithosphere = crust + rigid uppermost mantle

TRIGGER / MECHANISM

- Asthenosphere = weaker mechanical layer below
- ROCK-CYCLE NETWORK magma --crystallisation-→ IGNEOUS weathering + transport
- <---- melting -- METAMORPHIC <-- SEDIMENTARY heat + pressure compaction + cementation uplift and exposure reconnect all paths

EFFECT / INTERVENTION

- The cycle is a network: no rock family has one compulsory next stage.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: The crust is Earth's outer compositional shell, whereas the lithosphere is the stronger mechanical layer comprising the crust plus the rigid uppermost mantle.

01

STAGE 01 IGNEOUS, SEDIMENTARY AND METAMORPHIC DIAGNOSTIC MATRIX

IGNEOUS, SEDIMENTARY AND METAMORPHIC DIAGNOSTIC MATRIX | FIELD CLUES | MAGMA/LAVA COOLING | CRYSTALS, GLASS, VESICLES INTRUSIVE | SLOW COOLING | COARSE TEXTURE EXTRUSIVE | FAST COOLING | FINE OR GLASSY TEXTURE

FAMILY	PROCESS	FIELD CLUES
Sedimentary	deposit + lithify	beds, fossils, ripples
granite-rhyolite = felsic	diorite-andesite = intermediate gabbro-basalt = mafic	
FIREWALLS bedding is not foliation	loess and moraine are deposits until lithified porosity is void space	permeability is connected flow capacity

CLASSIFICATION

- FIELD CLUES
- magma/lava cooling
- crystals, glass, vesicles intrusive
- slow cooling
- coarse texture extrusive

DIAGNOSTIC BASIS

- fast cooling
- fine or glassy texture
- deposit + lithify
- beds, fossils, ripples
- solid-state change

PROCESS LINK

- grade, new minerals foliated
- differential stress
- aligned fabric
- Composition pairs
- granite-rhyolite = felsic

EXAM TRAP

- diorite-andesite = intermediate gabbro-basalt = mafic
- FIREWALLS bedding is not foliation
- loess and moraine are deposits until lithified porosity is void space
- permeability is connected flow capacity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Igneous rocks record the cooling history and composition of magma or lava: slow intrusive cooling favours coarse crystals, while rapid extrusive cooling favours fine, glassy or vesicular textures.

02

STAGE 02 FROM CONTINENTAL DRIFT TO PLATE TECTONICS AND WILSON-CYCLE CLOSURE

FROM CONTINENTAL DRIFT TO PLATE TECTONICS AND WILSON-CYCLE CLOSURE | WEGENER'S DRIFT CLAIM | SHELF-EDGE FIT | MATCHING ROCKS AND FOSSILS | LATER PALAEOMAGNETIC SUPPORT LIMITATION | NO ADEQUATE DRIVING MECHANISM | SEA-FLOOR SPREADING RIDGE CREATION | MAGNETIC STRIPES

WEGENER'S DRIFT CLAIM

shelf-edge fit → matching rocks and fossils → palaeoclimate → later palaeomagnetic support limitation: no adequate driving mechanism → SEA-FLOOR SPREADING ridge creation → magnetic stripes → older/deeper floor outward trench destruction

DEFINITION / COMPONENTS

- WEGENER'S DRIFT CLAIM
- shelf-edge fit → matching rocks and fossils
- palaeoclimate → later palaeomagnetic support limitation: no adequate driving mechanism

TRIGGER / MECHANISM

- SEA-FLOOR SPREADING ridge creation → magnetic stripes → older/deeper floor outward trench destruction
- PLATE TECTONICS divergence subduction transform creates crust destroys oceanic conserves crust ridge/rift trench + arc shallow quakes rift
- → narrow sea → ocean → subduction → collision → suture

EFFECT / INTERVENTION

- Wilson cycle is an analogy, not a compulsory destiny for every ocean.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plate boundaries are diagnosed by motion and products: divergence creates crust, subduction destroys oceanic crust, continent collision thickens crust, and transform motion conserves crust while generating shallow earthquakes.

- WEGENER'S DRIFT CLAIM
- shelf-edge fit → matching rocks and fossils
- palaeoclimate → later palaeomagnetic support limitation: no adequate driving mechanism

- SEA-FLOOR SPREADING ridge creation → magnetic stripes → older/deeper floor outward trench destruction
- PLATE TECTONICS divergence subduction transform creates crust destroys oceanic conserves crust ridge/rift trench + arc shallow quakes rift
- → narrow sea → ocean → subduction → collision → suture

- Wilson cycle is an analogy, not a compulsory destiny for every ocean.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plate boundaries are diagnosed by motion and products: divergence creates crust, subduction destroys oceanic crust, continent collision thickens crust, and transform motion conserves crust while generating shallow earthquakes.

03

STAGE 03 INDIAN PLATE JOURNEY AND THE THREE-PART GEOLOGICAL SKELETON

INDIAN PLATE JOURNEY AND THE THREE-PART GEOLOGICAL SKELETON | GONDWANA RIFTING | NORTHWARD INDIAN PLATE DRIFT | TETHYS CLOSURE | INDIA-EURASIA COLLISION ANCIENT PENINSULA HIMALAYAN FOLD-THRUST BELT FORELAND BASIN | NORTH-SOUTH EXAM CROSS-SECTION | SIWALIK HIMALAYA | FORELAND ALLUVIUM

Gondwana rifting → northward Indian Plate drift → Tethys closure → India-Eurasia collision ancient Peninsula Himalayan fold-thrust belt foreland basin cratons + belts shortening + thickening → stable framework continuing uplift/seismicity Indo-Ganga-Brahmaputra → NORTH-SOUTH EXAM CROSS-SECTION → Tethyan / Greater / Lesser / Siwalik Himalaya → foreland alluvium

SYSTEM / DEFINITION
- Gondwana rifting → northward Indian Plate drift → Tethys closure
- India-Eurasia collision ancient Peninsula Himalayan fold-thrust belt foreland basin cratons + belts shortening + thickening flexure + sediment
- stable framework continuing uplift/seismicity Indo-Ganga-Brahmaputra

PROCESS / MECHANISM
- NORTH-SOUTH EXAM CROSS-SECTION
- Tethyan / Greater / Lesser / Siwalik Himalaya → foreland alluvium
- Peninsular cratons, rifts, basins and Deccan cover

SPATIAL / INDIA PATTERN
- EVIDENCE LADDER superposition → cross-cutting → unconformity → fossil correlation
- radiometric event age → palaeomagnetism → geophysics
- Stable Peninsula means relatively stable, not uniform, flat or aseismic.

HAZARD / LIMIT / RESPONSE
- NORTH-SOUTH EXAM CROSS-SECTION

MECHANISM / VERDICT — Stable Peninsula means relatively stable, not uniform, flat or aseismic.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India's geological skeleton consists of an ancient Peninsular shield, an active Himalayan fold-thrust belt and a sediment-filled Indo-Ganga-Brahmaputra foreland basin linked by collision, flexure, erosion and deposition.

04

STAGE 04 INDIAN ROCK SYSTEMS, BASINS AND RESOURCE ASSOCIATIONS

INDIAN ROCK SYSTEMS, BASINS AND RESOURCE ASSOCIATIONS | INDIAN EXPRESSION | SAFE ASSOCIATION | GNEISS + SUPRACRUSTALS | PROCESS-SPECIFIC ORE | CUDDAPAH, VINDHYAN | STONE; BOUNDED ORES | FAULTED INLAND BASINS

INDIAN EXPRESSION → SAFE ASSOCIATION → gneiss + supracrustals → process-specific ore → Cuddapah, Vindhyan

CLASSIFICATION
- INDIAN EXPRESSION
- SAFE ASSOCIATION
- gneiss + supracrustals
- process-specific ore
- Cuddapah, Vindhyan

DIAGNOSTIC BASIS
- stone; bounded ores
- faulted inland basins
- coal, not universal
- Marine basins
- coasts and offshore

PROCESS LINK
- petroleum potential
- Deccan Trap
- stacked flood basalts
- soil, stone, water
- GONDWANA BASINS: Damodar

EXAM TRAP
- PETROLEUM SYSTEM
- source → maturation → migration → reservoir → seal → trap → timing
- DECCAN ARCHITECTURE
- flow top / vesicles / fractures / weathered contact / dense flow interior
- Association is not determinism: grade, preservation, exploration, technology, economics and environmental limits decide usable resources.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: source → maturation → migration → reservoir → seal → trap → timing.

05

STAGE 05 HIMALAYA, FORELAND, COASTS, ISLANDS AND STRUCTURAL CONTROLS

HIMALAYA, FORELAND, COASTS, ISLANDS AND STRUCTURAL CONTROLS | HIMALAYA COLLISION OROGEN | SUTURE FRAMEWORK | FORELAND FLEXURE AND FILL | BHANGAR/KHADAR ALLUVIAL EXPRESSIONS | PENINSULAR STRUCTURES | COASTS AND ISLANDS | NARMADA-SON AND TAPI RIFTS

HIMALAYA collision orogen: HFT/MFT → MBT → MCT → suture framework

FORELAND FLEXURE AND FILL → Bhabar → Terai → bhangar/khadar alluvial expressions → PENINSULAR STRUCTURES

SYSTEM / DEFINITION
- HIMALAYA collision orogen: HFT/MFT → MBT → MCT → suture framework
- FORELAND FLEXURE AND FILL
- Bhabar → Terai → bhangar/khadar alluvial expressions
- PENINSULAR STRUCTURES

PROCESS / MECHANISM
- COASTS AND ISLANDS
- Narmada-Son and Tapi rifts
- west: rifted-margin sectors
- Cambay and Kachchh

SPATIAL / INDIA PATTERN
- east: deltaic sediment basins
- Godavari rift basin
- Andaman: active island arc joints, faults, bedding
- Lakshadweep: coral atolls structure guides relief, drainage, waterfalls and groundwater climate + erosion + base level + human

HAZARD / LIMIT / RESPONSE
- action modify expression
- Rift = extensional zone
- graben = down-dropped block
- Lineament = mapped linear feature, not automatically an active fault.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: A rift is an extensional crustal zone, a graben is a down-dropped fault block, and a lineament is a mapped linear feature that is not automatically an active fault.

- FORELAND FLEXURE AND FILL
- Bhabar → Terai → bhangar/khadar alluvial expressions
- PENINSULAR STRUCTURES

- Narmada-Son and Tapi rifts
- west: rifted-margin sectors
- Cambay and Kachchh

- Godavari rift basin
- Andaman: active island arc joints, faults, bedding
- Lakshadweep: coral atolls structure guides relief, drainage, waterfalls and groundwater climate + erosion + base level + human

- Rift = extensional zone
- graben = down-dropped block
- Lineament = mapped linear feature, not automatically an active fault.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: A rift is an extensional crustal zone, a graben is a down-dropped fault block, and a lineament is a mapped linear feature that is not automatically an active fault.

06

STAGE 06 INDIA SEISMIC EXPRESSION AND GEOLOGY-TO-RISK TRANSLATION

INDIA SEISMIC EXPRESSION AND GEOLOGY-TO-RISK TRANSLATION | TECTONIC AND INHERITED STRUCTURE | NE ANDAMAN ARC KACHCHH | PENINSULA ACTIVE COLLISION SUBDUCTION INHERITED FAULTS/RIFTS STRONG SHAKING QUAKE | SITE FILTER | ALLUVIUM, BASIN DEPTH, GROUNDWATER, SLOPE, CONSTRUCTION | EXPOSURE + VULNERABILITY - CAPACITY | DISASTER RISK

TECTONIC AND INHERITED STRUCTURE

Himalaya / NE Andaman arc Kachchh / Peninsula active collision subduction inherited faults/rifts strong shaking → intraplate reactivation → SITE FILTER: alluvium, basin depth, groundwater, slope, construction

SYSTEM / DEFINITION
- TECTONIC AND INHERITED STRUCTURE
- Himalaya / NE Andaman arc Kachchh / Peninsula active collision subduction inherited faults/rifts strong shaking quake + tsunami
- intraplate reactivation

PROCESS / MECHANISM
- SITE FILTER: alluvium, basin depth, groundwater, slope, construction
- EXPOSURE + VULNERABILITY - CAPACITY
- DISASTER RISK

SPATIAL / INDIA PATTERN
- BIS Zones II-V are broad design generalisations, not predictions.
- Geological association must be followed by site evidence and engineering.
- RESPONSE: map → investigate → code → inspect → retrofit → prepare.

HAZARD / LIMIT / RESPONSE
- SITE FILTER: alluvium, basin depth, groundwater, slope, construction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India's seismicity reflects active Himalayan and Andaman plate boundaries plus reactivation of inherited structures such as Kachchh and selected Peninsular faults; seismic zoning is a design generalisation, not earthquake prediction.

07

STAGE 07 MAP-ANSWER AND MAINS SPINE FOR INDIA'S GEOLOGICAL ARCHITECTURE

MAP-ANSWER AND MAINS SPINE FOR INDIA'S GEOLOGICAL ARCHITECTURE | HOW DOES GEOLOGICAL STRUCTURE SHAPE INDIA'S SURFACE GEOGRAPHY? | 1. DEFINE | AGE + COMPOSITION + ROCK ARCHITECTURE + TECTONIC STRUCTURES | 2. DRAW | FORELAND PLAIN | PENINSULAR SHIELD CROSS-SECTION | 3. SEQUENCE

QUESTION: How does geological structure shape India's surface geography? → 1. DEFINE: age + composition + rock architecture + tectonic structures → 2. DRAW: Himalaya → foreland plain → Peninsular shield cross-section → 3. SEQUENCE: Gondwana rift → drift → collision → uplift → sediment fill → 4. MAP NAMED UNITS cratons

SYSTEM / DEFINITION
- QUESTION: How does geological structure shape India's surface geography?
- 1. DEFINE: age + composition + rock architecture + tectonic structures
- 2. DRAW: Himalaya → foreland plain → Peninsular shield cross-section

PROCESS / MECHANISM
- 3. SEQUENCE: Gondwana rift → drift → collision → uplift → sediment fill
- 4. MAP NAMED UNITS cratons
- Purana basins

SPATIAL / INDIA PATTERN
- Gondwana basins
- 5. LINK PROCESS TO CONSEQUENCE structure → relief/drainage/water/resource/seismic expression
- 6. QUALIFY association is not reserve; stable is not aseismic; alluvium has structure

HAZARD / LIMIT / RESPONSE
- 7. VERDICT inherited geology is filtered through tectonics, climate, erosion and use
- Answer order: thesis → map → evidence → mechanism → India example
- limitation → graded conclusion.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 7. VERDICT inherited geology is filtered through tectonics, climate, erosion and use.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | 47. OLDER INDIAN ROCK-SYSTEM SCHEMES | USEFUL ONLY WHEN DECLARED | 48. FIELD IDENTIFICATION AND METAMORPHIC NUANCE | 49. CRATONS, MOBILE BELTS AND ORE SYSTEMS | 50. BASIN ARCHITECTURE BEYOND THE NAME LIST | 51. DECCAN HYDROGEOLOGY AND LANDSCAPE REFINEMENT | 52. HIMALAYAN STRUCTURAL COMPLEXITY BEYOND A CLASSROOM SECTION

OPTIONAL PROCESS DEPTH
- 47. Older Indian rock-system schemes: useful only when declared
- 48. Field identification and metamorphic nuance
- 49. Cratons, mobile belts and ore systems
- 50. Basin architecture beyond the name list

DATA / INSTITUTION LIMIT
- 51. Deccan hydrogeology and landscape refinement
- 52. Himalayan structural complexity beyond a classroom section
- 53. Foreland, sediment routing and basin response
- 54. Hard-rock aquifers and engineering investigations

CONTEMPORARY PARALLEL
- 55. Mineral security without geological overclaim
- 56. Map and answer architecture
- Granite, diorite and gabbro provide coarse intrusive end-members; rhyolite, andesite and basalt are broad extrusive partners.
- Slate cleavage, schistosity and gneissic banding are fabrics, not sedimentary beds.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels